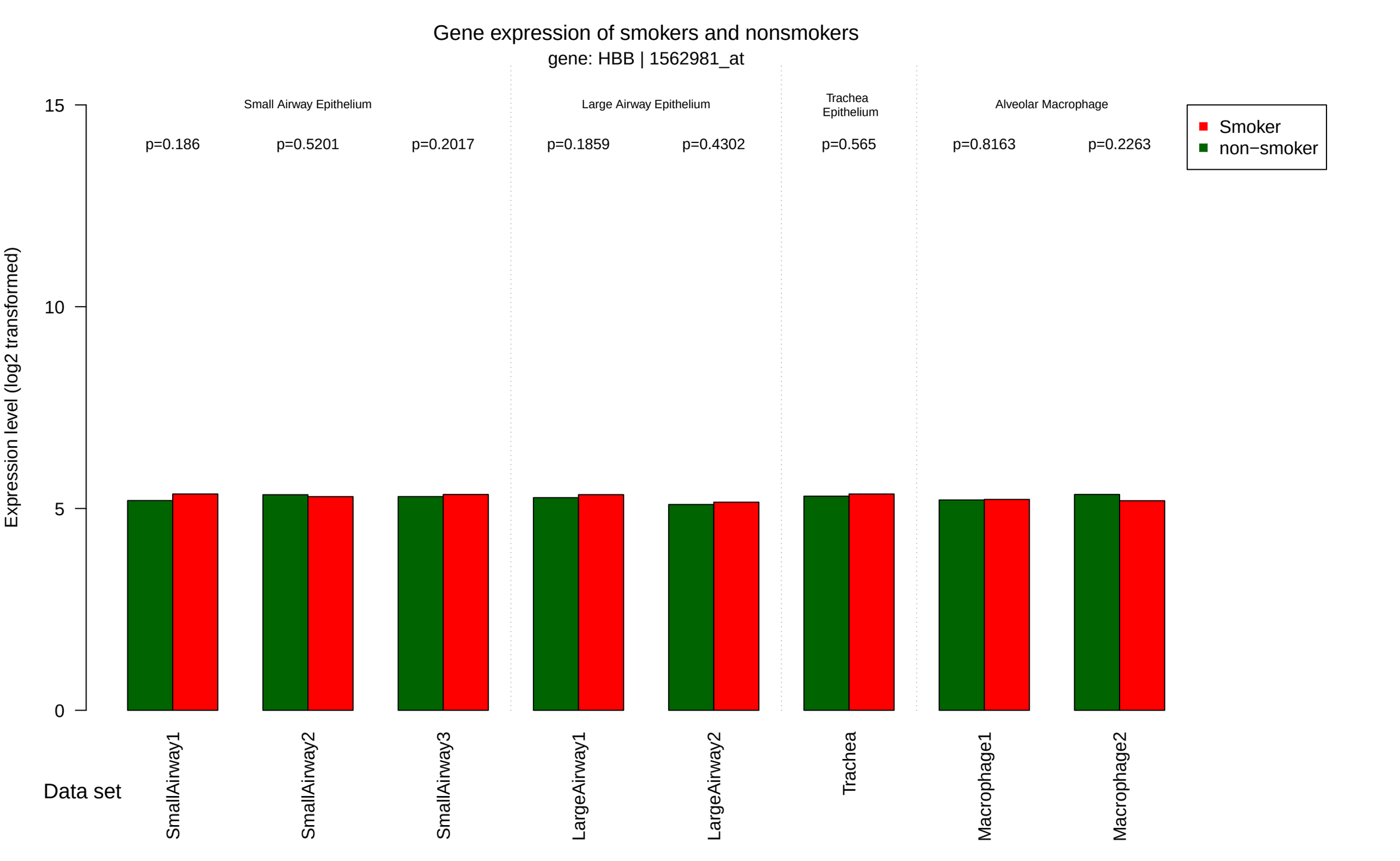
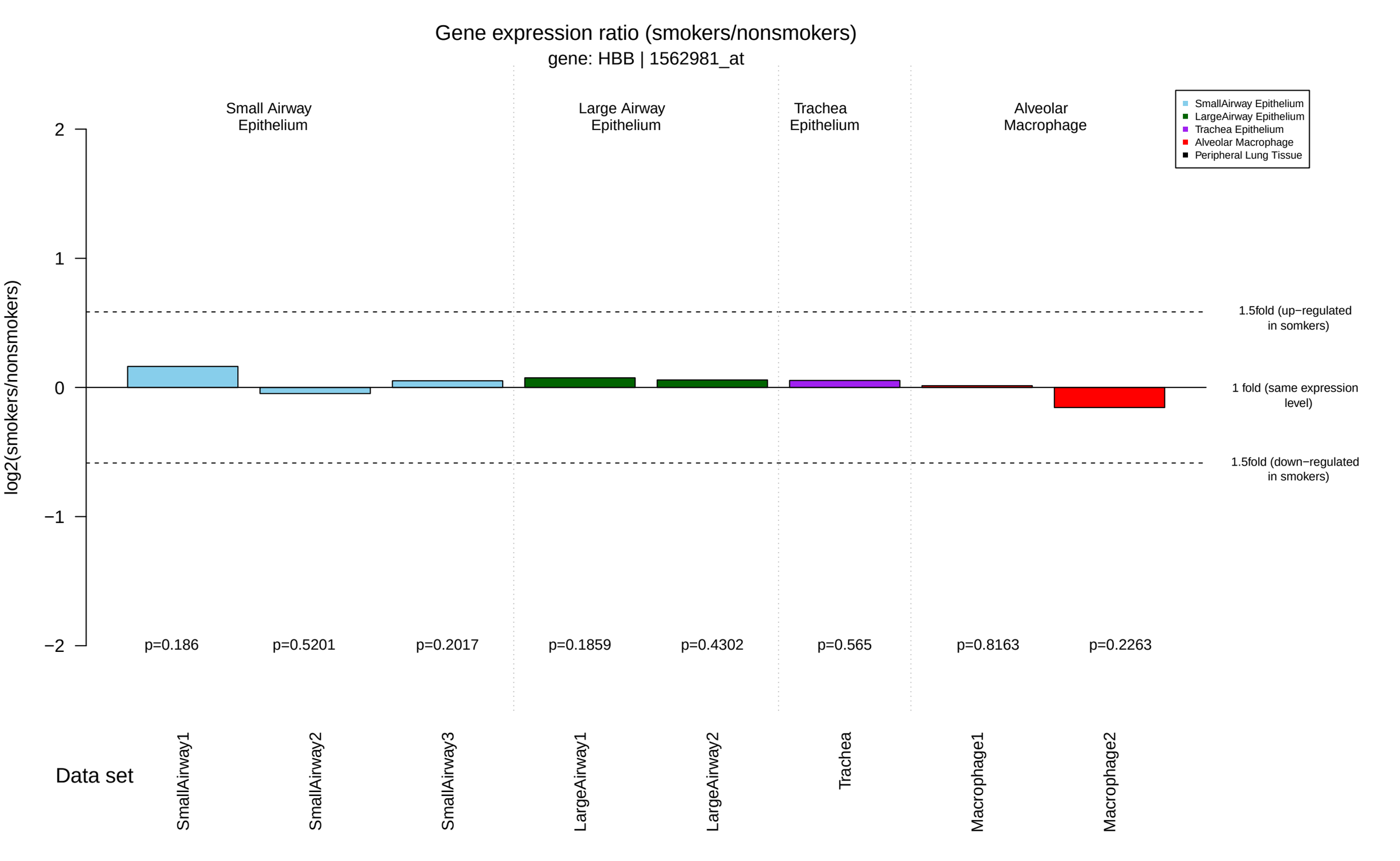
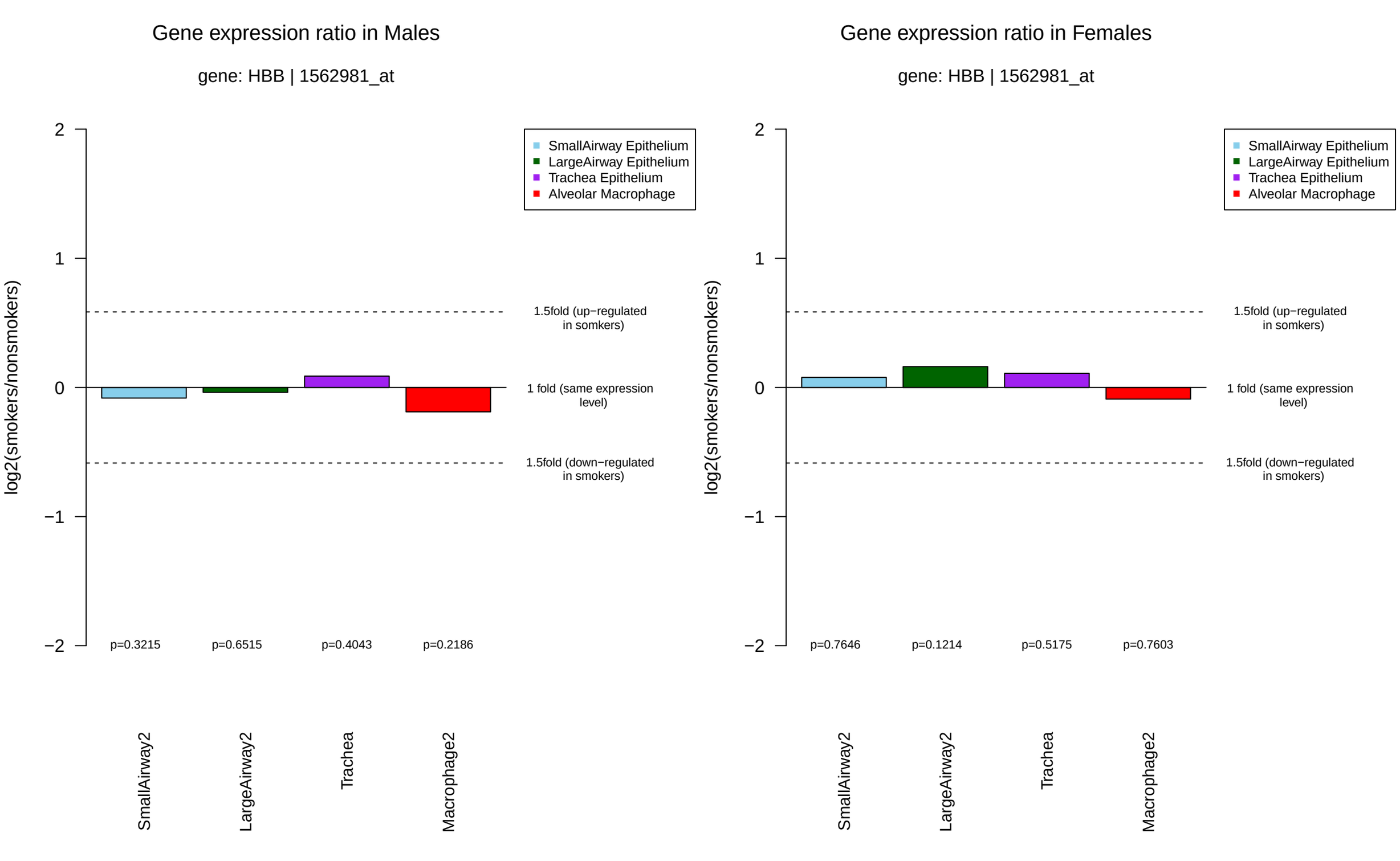
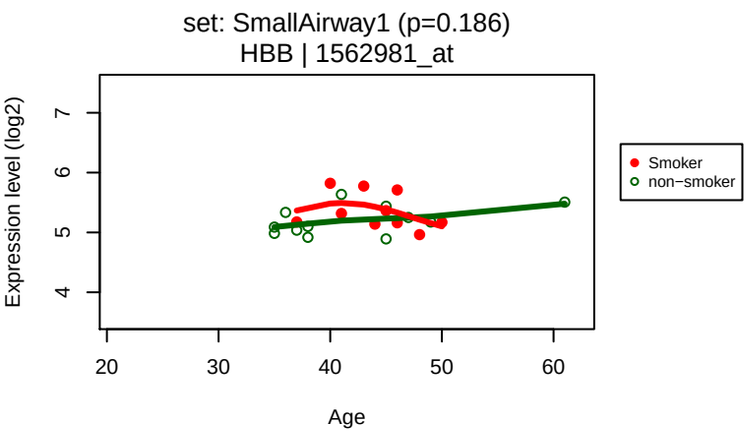
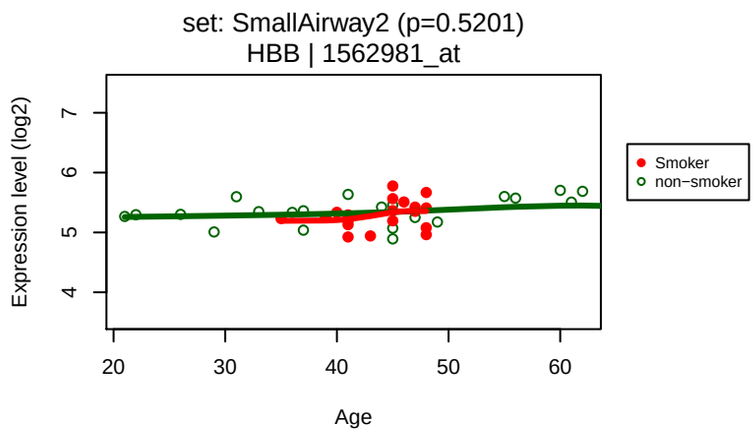
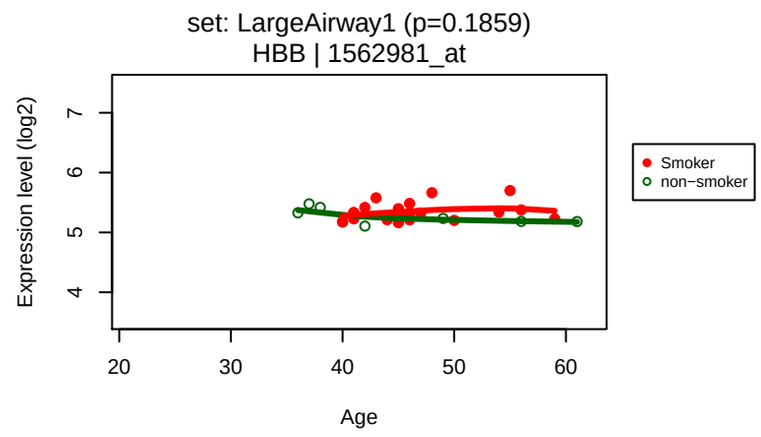

Gene expression of smokers and nonsmokers
gene: HBB | 1562981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.186
p=0.5201
p=0.2017
p=0.1859
p=0.4302
p=0.565
p=0.8163
p=0.2263
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

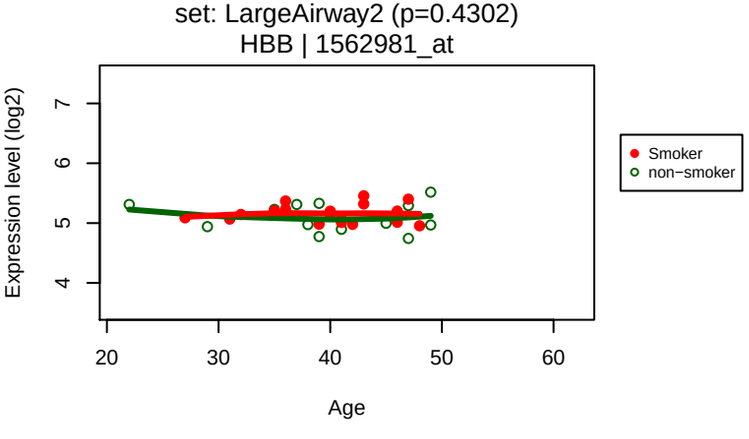

Gene expression ratio (smokers/nonsmokers)
gene: HBB | 1562981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.186
p=0.5201
p=0.2017
p=0.1859
p=0.4302
p=0.565
p=0.8163
p=0.2263
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

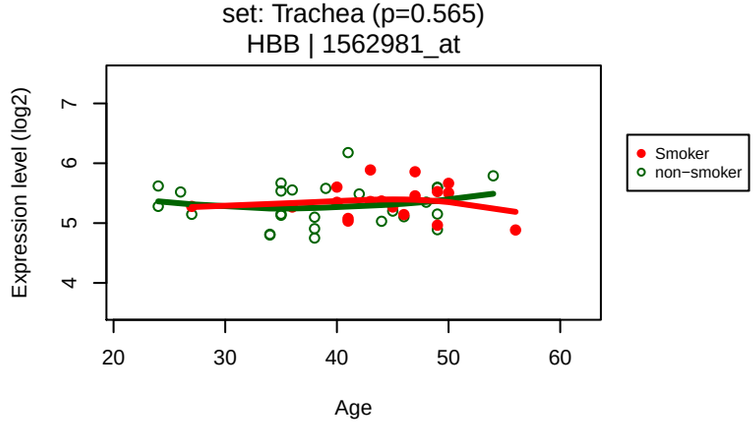

Gene expression ratio in Males
gene: HBB | 1562981_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3215
p=0.6515
p=0.4043
p=0.2186
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HBB | 1562981_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7646
p=0.1214
p=0.5175
p=0.7603
SmallAirway2
LargeAirway2
Trachea
Macrophage2

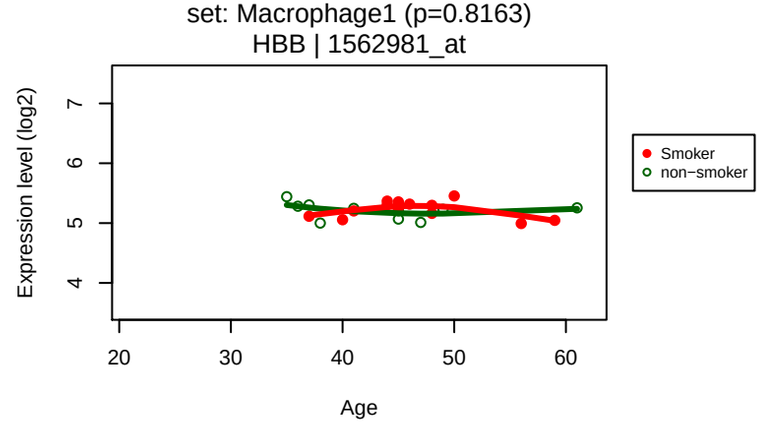

set: SmallAirway1 (p=0.186)
HBB | 1562981_at
Expression level (log2)
Age
Smoker
non−smoker

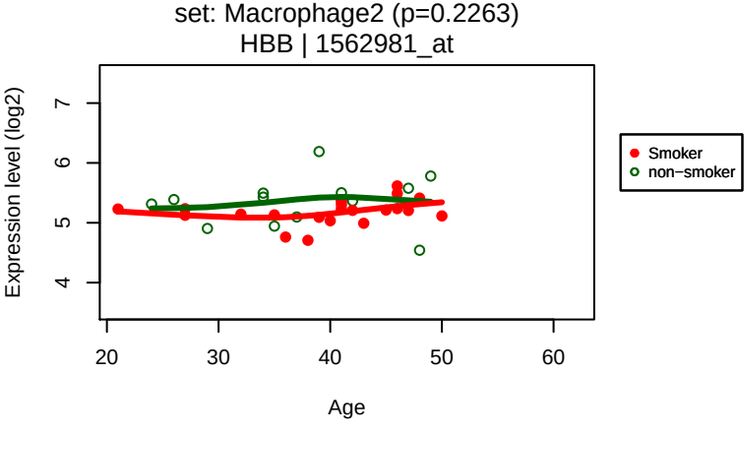

set: SmallAirway2 (p=0.5201)
HBB | 1562981_at
Expression level (log2)
Age
Smoker
non−smoker

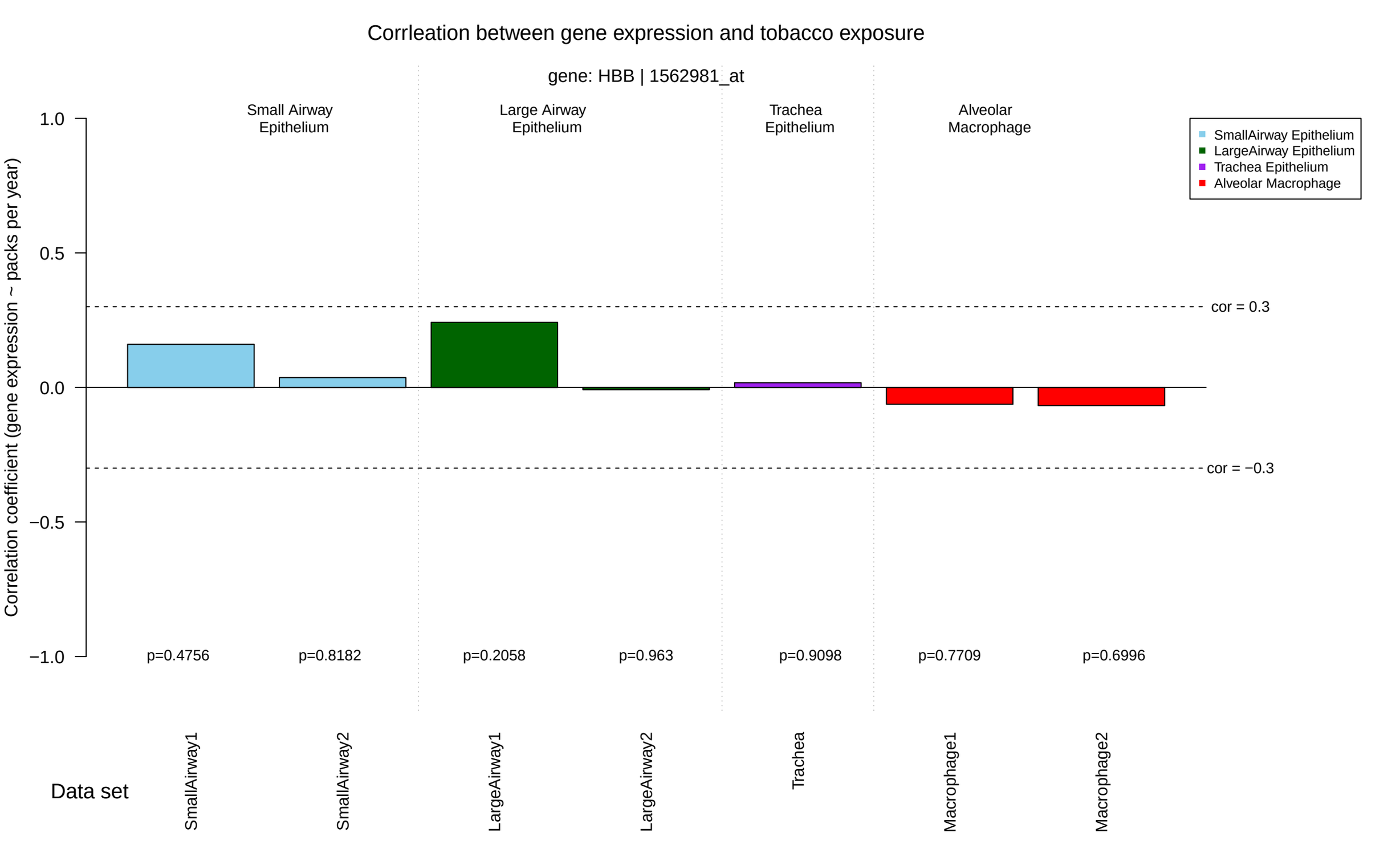

set: LargeAirway1 (p=0.1859)
HBB | 1562981_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4302)
HBB | 1562981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.565)
HBB | 1562981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8163)
HBB | 1562981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2263)
HBB | 1562981_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HBB | 1562981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4756
p=0.8182
p=0.2058
p=0.963
p=0.9098
p=0.7709
p=0.6996
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2